Look Where You Work: View-Aligned Asset Identification with UWB and Smart Glasses

1st Mathias Pelka
Technische Hochschule Lübeck
Lübeck, Germany
mathias.pelka@th-luebeck.de

2nd Thomas Willemsen
Jade Hochschule
Oldenburg, Germany
willemsen@jade-hs.de

Abstract—In dense industrial workspaces, operators using smart glasses often need asset-specific information for the machine they are currently observing. This is difficult because indoor positioning alone does not reliably determine the relevant asset when several nearby machines remain plausible candidates. As a result, the system may know the operator position, but not which workstation is actually being attended to. To address this limitation, we present a low-cost approach that combines Ultra Wideband (UWB) positioning with on-device head-orientation sensing on commodity Android smart glasses. After a short heading calibration, the system fuses position and viewing direction to infer the most likely observed workstation and retrieves the corresponding device data via MQTT. We evaluate the approach in a controlled 3D-printer laboratory using a Leica laser-tracker reference system, static and dynamic motion experiments, and an exploratory practical identification test. The results show that UWB provides coarse spatial context but does not reliably disambiguate nearby assets on its own. They further show that orientation-based disambiguation is feasible in principle, but depends strongly on the device and becomes unreliable during very slow rotations. Within the evaluated laboratory scenario, the combined approach nevertheless enabled correct workstation identification in the tested cases.

Index Terms—Ultra-Wideband (UWB), Augmented Reality (AR), Smart Factory, Industrial IoT, Context-Aware Interfaces, Human-Machine Interaction, Location-Based Services

I. Introduction

Modern shop floors in smart factories are increasingly instrumented, yet many relevant tasks still take place under hands-busy conditions, for example while carrying parts, handling tools, or working in constrained spaces. In such situations, operators may need asset-specific information for the machine they are currently observing without interrupting their work to navigate menus or manually select a device. Smart glasses are a promising interface for this purpose because they provide hands-free information directly in the user's field of view [1]. However, indoor positioning alone is often insufficient in dense layouts, since user position does not indicate which nearby asset is currently being observed. Reliable context selection therefore requires an additional spatial cue such as head orientation or viewing direction [2].

A practical solution is to combine smart glasses with indoor positioning in a low-cost architecture [3] and to extend this setup with gyroscope-based head-orientation sensing. Ultra-wideband (UWB) is a well-established technology for indoor localization in industrial environments [4]–[6], while gyroscopes provide a practical signal for estimating head rotation on wearable devices, although drift and device-dependent stability remain relevant challenges [7]. By combining position and head orientation, asset selection can be aligned not only with the worker location but also with the direction the user is facing.

This paper makes the following contributions:

- It identifies asset disambiguation in dense indoor industrial layouts as a limitation of position-only AR context selection and motivates head orientation as an additional spatial cue.
- It presents a low-cost smart-glasses architecture that combines UWB positioning, on-device orientation sensing, and MQTT-based integration for workstation-level context inference.
- It provides a reference-based evaluation showing that UWB supports coarse spatial context but not reliable asset disambiguation on its own, while orientation-based selection is feasible but strongly affected by device-dependent behavior and slow rotations.

The remainder of the paper is structured as follows. Section II reviews related work. Section III outlines the system architecture. Section IV describes the smart-glasses application, and Section V presents pose estimation and context inference. Section VI introduces the laboratory geometry. Section VII reports the evaluation results, and Section VIII concludes the paper.

II. Related Work

This work relates to industrial AR assistance with smart glasses, indoor localization with UWB, and head-orientation-based target selection. Our contribution lies at their intersection, with a focus on asset disambiguation in industrial layouts.

Smart glasses and AR have been studied as hands-free interfaces for industrial assistance, including task guidance, maintenance support, and mobile information access at the point of work [1], [8]. Reviews further emphasize the importance of spatial context for effective AR support in smart-industry settings [9]. Flatt et al.

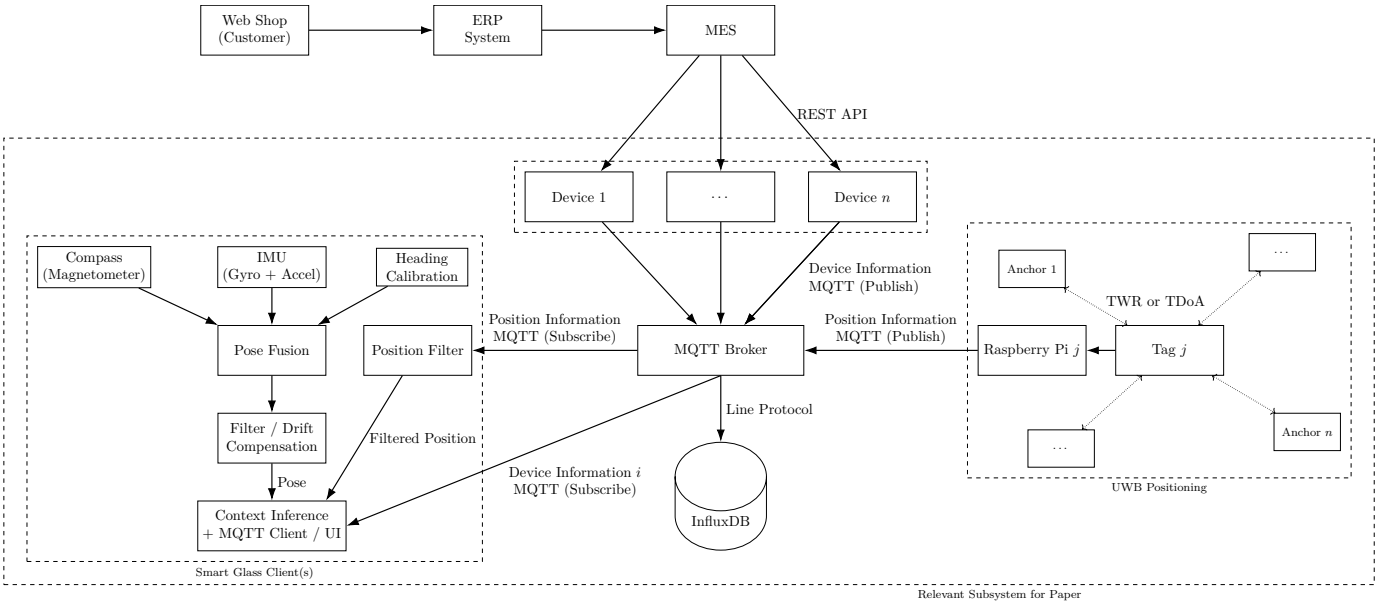


Fig. 1: System overview. Orders are processed by ERP and MES and assigned to shop-floor devices, which publish status data via MQTT. A UWB setup provides operator position estimates. The smart-glasses app combines position and calibrated head orientation for workstation-level context inference and renders the selected device information in the AR interface.

[10] combine AR and indoor localization for maintenance support, but do not address asset disambiguation.

UWB is a well-established technology for precise indoor positioning in industrial environments. Prior work has examined ranging methods and non-line-of-sight effects [4], [5], and recent reviews confirm its relevance for Industry 4.0 scenarios [6]. UWB has also been combined with head-mounted displays for AR visualization [11]. However, this line of work mainly focuses on localization accuracy and robustness rather than on selecting the correct asset in dense layouts.

Research on inertial sensing and head-based interaction shows that IMU-based pose estimation is useful but affected by drift and device-dependent behavior [7]. UWB and IMU have often been fused to improve localization robustness [12], and gyroscope-based head orientation has been explored as a cue for target selection on head-mounted displays [13]. These works, however, usually treat localization and interaction separately.

Our earlier work, Step Into Context [3], introduced a low-cost architecture for location-aware AR feedback based on UWB and MQTT, but relied on proximity-based context selection. While effective in sparse settings, that approach becomes ambiguous when devices are close to each other.

Overall, prior work establishes the value of smart glasses for industrial assistance, UWB for indoor positioning, and head orientation for target selection. What remains insufficiently addressed is their combination for low-cost asset disambiguation in dense industrial indoor environments. This paper addresses that gap by combining UWB position and head orientation for workstation-level context inference on commodity smart glasses.

III. System Overview

Our system extends the low-cost architecture of our earlier work [3] with orientation-aware context selection on smart glasses. Figure 1 summarizes the relevant data flow. In the production environment, jobs are assigned to shop-floor devices via the existing ERP and MES infrastructure. These devices publish telemetry data via MQTT, which makes live machine information available to subscribing clients.

In parallel, a UWB subsystem provides spatial information for mobile entities. A Raspberry Pi connected to a Qorvo DWM1001 publishes position estimates to the same MQTT bus.

On the operator side, a unified Android application runs on smart glasses or other Android-based wearable devices. The app subscribes to both UWB position data and device telemetry. After a brief user-triggered calibration, it estimates head orientation from the on-device inertial sensors and uses relative heading updates to reduce sensitivity to magnetic disturbances. Incoming UWB positions are filtered to suppress jitter and outliers.

A context-inference module then combines filtered position and head orientation to determine the most likely workstation the operator is facing. Based on this result, the application selects the corresponding device stream and renders the relevant information in a view-aligned AR

Fig. 2: Smart glasses used in the laboratory setup. The Vuzix Blade 2 (top), the Google Glass Enterprise Edition 2 (middle), and the Vuzix M4000 (bottom).

interface. A reset function allows the heading reference to be re-initialized when drift becomes excessive.

IV. Smart Glasses

In industrial environments, smart glasses act as hands-free user interfaces that present context-dependent instructions, live video, and machine data in the operator's field of view.

A single Android application is deployed on Google Glass Enterprise Edition 2, Vuzix M4000, and Vuzix Blade 2. For comparison, the same application is also executed on a Google Pixel 8 Pro as a conventional smartphone baseline. Table I summarizes selected device characteristics relevant to deployment.

All three devices provide inertial orientation sensing based on a 3-axis gyroscope, accelerometer, and magnetometer. However, official vendor documentation does not report gyroscope quality metrics such as drift, bias stability, noise density, or calibration accuracy. A device-level comparison of gyroscope quality therefore cannot be derived from documentation alone and must be assessed experimentally. Although these devices are older models, direct contemporary replacements remain limited, particularly for Android-based smart glasses with integrated displays suitable for custom AR applications.

V. Pose Estimation and 2D Context Inference

The smart-glasses application estimates the wearer pose using the Android sensor framework. Orientation is obtained from the fused rotation estimate provided by the Android SensorManager, based on inertial measurements from the gyroscope and accelerometer. The yaw angle is used as the primary output, since workstation disambiguation is performed in a 2D floor-plane model.

UWB positioning data are received via MQTT and filtered on-device to suppress outliers and jitter. For this purpose, an α-trimmed filter is applied over a sliding window of recent position samples by discarding the largest and smallest α fraction and averaging the remaining values. This yields a robust position estimate while preserving low latency.

The spatial coordinates of all candidate assets are stored in the application configuration and loaded at runtime. On application startup, the user performs a short heading calibration by facing a predefined reference direction. This defines the offset between the device-reported yaw and the global coordinate frame used for the device map.

Given the filtered user position $\mathbf{p} \in \mathbb{R}^2$ and the calibrated heading θ, context inference selects the most likely observed device in 2D. For each device with known position $\mathbf{d}_i \in \mathbb{R}^2$, the bearing angle is computed as

$$\phi_i = \operatorname{atan2}((\mathbf{d}_i - \mathbf{p})_y, (\mathbf{d}_i - \mathbf{p})_x)\,,$$

with wrapped angular difference

$$\Delta_i = \operatorname{wrapToPi}(\phi_i - \theta).$$

The selected device is the candidate with minimal absolute angular deviation $|\Delta_i|$ within a fixed field-of-view threshold $\Delta_{\max}$. In the rare case of equal angular deviation, the nearer device is selected.

VI. Context Inference in the Laboratory Geometry

Figure 3 shows the laboratory environment used in the experiments. Four fixed reference points, labeled A to D, define the workspace geometry. The evaluation considers four 3D printers, denoted as p_A to p_D. The printers p_A, p_B, and p_C are arranged in one row, while p_D is placed separately. Additional equipment such as robots and conveyor belts is present in the lab but excluded from the evaluation. Each printer is represented by the center of its footprint in a 2D floor-plane model, with coordinates estimated using a laser tracker.

Figures 3a and 3b illustrate the two main geometric cases of the proposed context inference. In the unambiguous case, the viewing direction leads to a single dominant candidate, so the minimum-angle rule selects p_A directly. In the ambiguous case, several devices lie within the field of view and show similar angular deviations. The system then resolves the ambiguity by selecting the nearest candidate, which again yields p_A.

VII. Evaluation

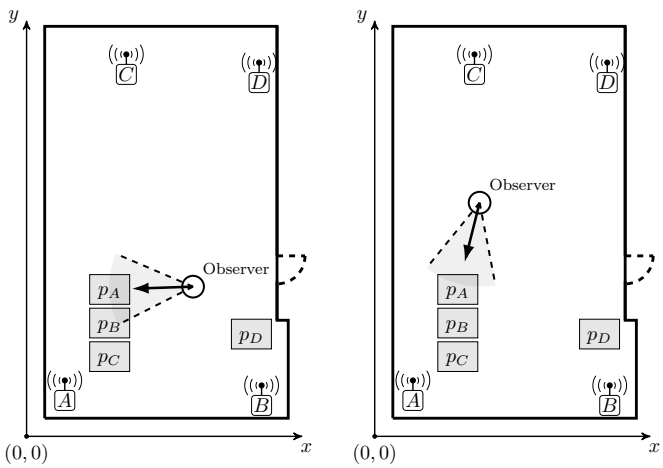
This section evaluates the proposed system from six perspectives: experimental setup and reference instrumentation, UWB position accuracy, static orientation stability, rotational-motion response, dynamic 2D trajectory behavior, and an exploratory practical test for object identification in the laboratory scenario.

A. Experimental Setup

To evaluate the proposed system under identical motion conditions, a custom multi-device holder (Fig. 4) was fabricated using a 3D printer. It allowed three smart glasses and one smartphone to be mounted simultaneously, so that

TABLE I: Comparison of smart glasses considered for the system integration. Values are based on official vendor documentation.

Property	Google Glass EE2	Vuzix M4000	Vuzix Blade 2
Operating system	Android 8.1 (AOSP)	Android 11	Android 11
SoC / processor	Qualcomm XR1	Qualcomm XR1	Qualcomm Snapdragon 429
Display resolution	640×360	854×480	480×480
Display technology	Prism / see-through	Waveguide / see-through	Waveguide / see-through
Field of view (display, diagonal)	n/a	28°	20°
Camera	8 MP	12.8 MP	8 MP
Memory / storage	3 GB / 32 GB	6 GB / 64 GB	1 GB / 40 GB
Industrial wear / safety	clips to glasses / safety shields	safety helmet mounts available	ANSI Z87.1 certified



(a) Unambiguous selection. (b) Ambiguous case.

Fig. 3: 2D context inference for device selection.

Fig. 4: Custom 3D-printed holder for the experiments

all devices experienced the same translation and rotation during the experiments.

The holder also contained the UWB tag used by the indoor positioning system. In addition, a 6DOF T-Probe was attached to the front of the holder and tracked by a Leica laser tracker, which served as external reference for position and orientation.

Static stability, rotational-motion, and dynamic trajectory tests were carried out in the laboratory. During all experiments, the complete holder was moved along predefined paths so that all smart glasses, the smartphone, and the UWB tag followed exactly the same motion. This enabled a direct comparison of the recorded signals under controlled but realistic conditions.

The UWB system operated at 3 Hz. All smart-glasses data and reference data from the Leica tracker were recorded with timestamps and temporally aligned in post-processing using cross-correlation. Specifically, the cross-correlation was calculated between the yaw-angle time series of the smart glasses and the laser tracker in order to estimate the temporal offset. The lag corresponding to the maximum correlation was used as the synchronization shift between both time series. Overall, the setup provided a repeatable basis for comparing multiple wearable devices against a common reference.

B. Reference System and UWB Position Accuracy

The reference system used in this work consists of a Leica Absolute Tracker AT960 with a 6DOF T-Probe [16]. As a high-accuracy optical 3D measurement system, it serves as an external reference for the lower-rate smart-glasses and UWB measurements [15]. The sampling rate was set to 30 Hz.

For the experiments, the laser tracker was registered to the UWB coordinate system using four access points measured with CornerCube reflectors. The resulting coordinate transformation had a residual standard deviation below 10 mm.

To assess UWB position accuracy, eight measurement points were evaluated in the transformed coordinate system by comparing the UWB estimates with the corresponding laser-tracker reference coordinates.

The results show that the reference measurements in the UWB coordinate system are accurate to better than 10 mm. Across the selected test points, the UWB system achieved a mean 2D position error of 0.17 m, consistent with our previous work. This is sufficient for coarse spatial context, but not for reliable asset disambiguation in dense layouts where several nearby devices may remain plausible candidates [3].

C. Static Stability

To evaluate the stability of the orientation estimates in a static setting, we performed a continuous measurement over 5 min while the setup remained at rest. The recorded orientation signals were compared against the

Leica reference system, and the deviations in yaw, pitch, and roll were analyzed. The resulting standard deviations are summarized in Table II.

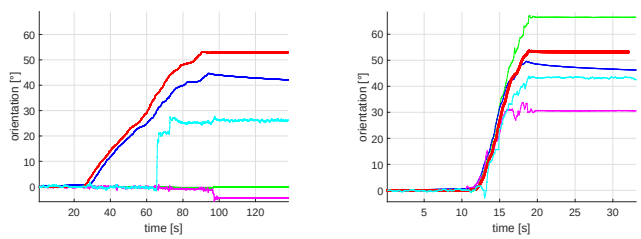
TABLE II: Standard deviation of the orientation error during a static 5 min measurement. Values are given in degrees.

Device	Yaw	Pitch	Roll
Google Glass EE2	0.0237	0.0095	0.0070
Pixel 8	0.1670	0.0065	0.0054
Vuzix Blade 2	0.0706	0.0696	0.0142
Vuzix M4000	0.3466	0.2942	0.0781

Overall, Table II shows clear device-dependent differences in static orientation stability. Google Glass was most stable, whereas the Vuzix M4000 showed the largest variation relative to the reference system.

D. Sensitivity to Rotational Motion

This experiment examined how the evaluated devices respond to rotational motion. For the proposed application, slow rotations are particularly relevant, since operators often turn their heads gradually when inspecting nearby assets. In low-cost sensor systems, however, such motion may be suppressed or insufficiently integrated by device-side sensor fusion. To show this, two experiments with different angular velocities were carried out.



(a) Slow rotation. (b) Fast rotation.

Fig. 5: Sensitivity to rotational motion for slow and fast rotations. Color coding: green = Google Glass, blue = Pixel 8, magenta = Vuzix Blade 2, cyan = Vuzix M4000, and red = laser-tracker reference.

Figure 5 summarizes the results. In the first experiment, the setup was rotated slowly by about 55° over 65 s. As shown in Fig. 5a, Google Glass and the Vuzix Blade 2 hardly detected this motion at all, since both signals remained nearly constant despite the continuous change in the reference orientation. The Pixel 8 captured part of the rotation, but clearly underestimated the total angle. The Vuzix M4000 did not follow the reference continuously either and instead showed an abrupt change later in the measurement.

In the second experiment, a similar rotation of about 54° was performed within 8 s. As shown in Fig. 5b, all devices reacted more clearly to the motion. Google Glass followed the rotation much better than in the slow case, but overestimated the final angle by about 13°. The Pixel 8 again tracked the motion reasonably well, but still with a noticeable deviation from the reference. The Vuzix Blade 2 and the Vuzix M4000 also responded more strongly than before, but both remained substantially below the reference trajectory.

Position remained approximately constant in both experiments, so the observed differences can be attributed mainly to orientation estimation.

Overall, slow rotations posed a clear problem for all evaluated devices. Faster rotations were detected more reliably, but noticeable errors remained. For practical context inference, this shows that device suitability depends strongly on how well slow head movements are captured.

E. Dynamic 2D Trajectory Experiment

In this experiment, the test body was moved through the measurement area on a cart and repeatedly rotated around the vertical axis in both directions.

As in the previous experiments, the data were synchronized by cross-correlation for direct comparison with the reference data. Figure 6 shows the UWB-based position estimates and the laser-tracker reference trajectory. The standard deviation of the position differences was 0.47 m in X and 0.66 m in Y. Compared with the static point measurements, the larger deviations can be attributed to the dynamic motion of the tag and temporary shadowing during the cart movement. The results confirm that UWB supports coarse localization, but is not sufficient for reliable disambiguation of closely spaced devices.

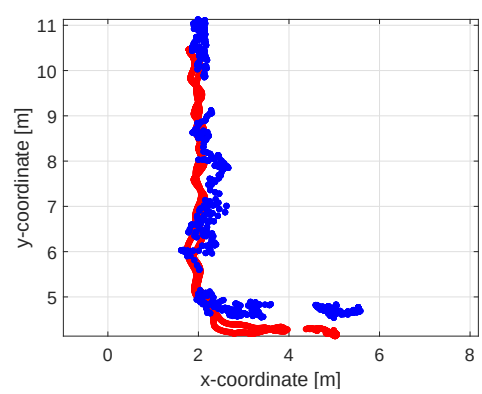


Fig. 6: UWB trajectory and laser-tracker reference during dynamic motion.

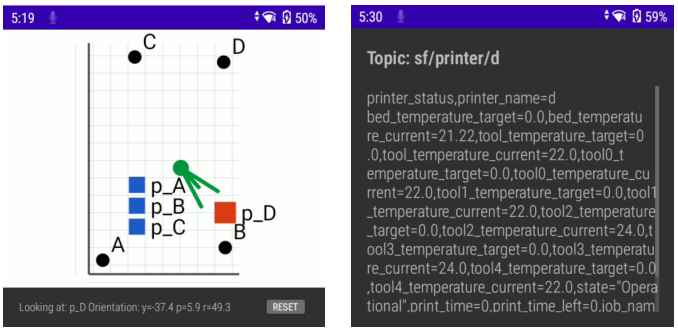
The figure shows that the overall trajectory was captured, but visible deviations from the reference path remained. The dynamic experiment primarily assessed positional behavior under motion. For the Vuzix devices, unresolved alignment issues between device orientation and reference frame currently prevent a reliable angular comparison with the laser-tracker data.

F. Exploratory Practical Test

In addition to the technical evaluation, an exploratory practical test was conducted in the laboratory. The test

combined the UWB positioning system with the smart glasses to identify four 3D printers. Several test positions were selected, and the user looked at the corresponding devices from each position.

For all three evaluated smart glasses, namely the Google Glass Enterprise Edition 2, the Vuzix Blade 2, and the Vuzix M4000, the system correctly identified the observed printer in the tested situations. Figure 7 shows one representative case.



(a) Main view during device selection.

(b) Detailed view of printer D.

Fig. 7: Smart-glass application during the practical test.

The results indicate feasibility under laboratory conditions. However, the test remains an initial proof-of-concept and does not establish general robustness.

VIII. Conclusion and Outlook

This work showed that low-cost UWB positioning and on-device orientation sensing can be combined for context-aware device identification in a laboratory environment. The UWB system achieved a dynamic mean 2D position error of 0.66 m, which is sufficient for coarse localization but not for reliable asset disambiguation in dense layouts on its own. The evaluation also showed that very slow rotations are tracked unreliably and that orientation behavior differs across devices. In addition, unresolved orientation-frame alignment issues prevented a reference-based angular evaluation for the Vuzix devices. In an exploratory practical test, all evaluated smart glasses correctly identified the viewed device in the tested situations. Future work will include a structured user study and a closer analysis of rotational-speed requirements as well as device-specific alignment and calibration effects.

Open-Source Artifact

We release the complete Android reference implementation and the measurement data required to reproduce the core results: github.com/thl-smart-factory-lab/look-where-you-work.

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